Object Oriented Programming (CSEG1044)

Unit 06 – Lecture 03: UML Class Design + Package Structure

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February 27, 2026

Sources: Git SCM: *Pro Git* (2e) (version control workflow), accessed February 27, 2026.; GitHub Docs (repositories, issues, pull requests), accessed February 27, 2026.; Oracle Java Tutorials: Swing + JDBC (accessed February 27, 2026).; JUnit 5 User Guide (accessed February 27, 2026).; Apache Maven Project: Getting Started (accessed February 27, 2026).; Gradle User Manual: Getting Started (accessed February 27, 2026).; UML class diagrams: UML 2.x overview/spec (accessed February 27, 2026).

Quick Links

Overview Key Topics Demo Summary

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Agenda

1 Overview

2 Key Topics

3 Demo / Example

4 Summary

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Learning Outcomes

- Convert use cases into classes and responsibilities.

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Learning Outcomes

- Convert use cases into classes and responsibilities.
- Reduce coupling using interfaces and layered design.

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Learning Outcomes

- Convert use cases into classes and responsibilities.
- Reduce coupling using interfaces and layered design.
- Complete Deliverable-2: class diagram draft + package structure.

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Unit 06: Capstone Project

- Lecture focus: UML Class Design + Package Structure

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Today: Roadmap

- Identify classes, relationships, responsibilities

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Today: Roadmap

- Identify classes, relationships, responsibilities
- Interfaces/abstractions to reduce coupling

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Today: Roadmap

- Identify classes, relationships, responsibilities
- Interfaces/abstractions to reduce coupling
- Deliverable-2: class diagram (draft) + package/module structure

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UML notation (quick cheat sheet)

- Class box: **Name** / **Attributes** / **Methods**

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UML notation (quick cheat sheet)

- Class box: **Name** / **Attributes** / **Methods**
- Visibility: + public, - private, # protected
- Relationships: association (line), inheritance (triangle), composition (filled diamond)
- Multiplicity: `1`, `0..1`, `*`, `1..*`

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From use cases to classes

1. Use cases → nouns (entities) + verbs (operations)

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From use cases to classes

1. Use cases → nouns (entities) + verbs (operations)
2. Entities: Resource, Booking, User

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From use cases to classes

1. Use cases → nouns (entities) + verbs (operations)
2. Entities: Resource, Booking, User
3. Services: BookingService (rules)

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From use cases to classes

1. Use cases → nouns (entities) + verbs (operations)
2. Entities: Resource, Booking, User
3. Services: BookingService (rules)
4. DAOs: BookingDao (DB)

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From use cases to classes

1. Use cases → nouns (entities) + verbs (operations)
2. Entities: Resource, Booking, User
3. Services: BookingService (rules)
4. DAOs: BookingDao (DB)
5. UI: panels/screens

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Responsibilities (avoid God classes)

- UI: read inputs, show messages

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Responsibilities (avoid God classes)

- UI: read inputs, show messages
- Service: validate rules (overlap checks)

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Responsibilities (avoid God classes)

- UI: read inputs, show messages
- Service: validate rules (overlap checks)
- DAO: SQL + CRUD only

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Relationships (use carefully)

- Prefer composition (has-a) for most domain links.

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Relationships (use carefully)

- Prefer composition (has-a) for most domain links.
- Use inheritance only for true is-a.

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Relationships (use carefully)

- Prefer composition (has-a) for most domain links.
- Use inheritance only for true is-a.
- Interfaces help reduce coupling (e.g., repository/DAO interface).

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Package structure (recommended)

- `com.upes.crm.model`

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Package structure (recommended)

- `com.upes.crm.model`
- `com.upes.crm.dao`

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- `com.upes.crm.model`
- `com.upes.crm.dao`
- `com.upes.crm.service`

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Package structure (recommended)

- `com.upes.crm.model`
- `com.upes.crm.dao`
- `com.upes.crm.service`
- `com.upes.crm.ui`

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- `com.upes.crm.dao`
- `com.upes.crm.service`
- `com.upes.crm.ui`
- `com.upes.crm.util`

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Deliverable-2 checklist

- UML class diagram draft (entities + service + DAO + UI)

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Deliverable-2 checklist

- UML class diagram draft (entities + service + DAO + UI)
- Key relationships shown (Booking → Resource)

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Deliverable-2 checklist

- UML class diagram draft (entities + service + DAO + UI)
- Key relationships shown (Booking → Resource)
- Package structure documented (tree or list)

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Deliverable-2 checklist

- UML class diagram draft (entities + service + DAO + UI)
- Key relationships shown (Booking → Resource)
- Package structure documented (tree or list)
- Initial skeleton folders/packages committed (optional)

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In-class Activity (evidence)

- Draft class diagram uploaded to repo

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In-class Activity (evidence)

- Draft class diagram uploaded to repo
- Package structure committed (skeleton)

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In-class Activity (evidence)

- Draft class diagram uploaded to repo
- Package structure committed (skeleton)
- Issues assigned per module (UI/DB/service)

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Key Takeaways

- Design comes from use cases, not from guessing.

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Key Takeaways

- Design comes from use cases, not from guessing.
- Layering reduces coupling and makes changes easier.

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Key Takeaways

- Design comes from use cases, not from guessing.
- Layering reduces coupling and makes changes easier.
- Deliverable-2 unlocks implementation work.

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Checkpoint Questions

1. Why separate UI/service/DAO?

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Checkpoint Questions

1. Why separate UI/service/DAO?
2. Give one example of composition in your project.

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Checkpoint Questions

1. Why separate UI/service/DAO?
2. Give one example of composition in your project.
3. What packages will your project contain?

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Next Lecture Preview

Next: Database Design + JDBC Layer

- Convert entity model to tables and implement DAO CRUD.

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